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Experimental performance evaluation and modeling of jet impingement cooling for thermal management of photovoltaics

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ABSTRACT

Thermal management of PV systems is one of the most important factors that affect the overall performance especially in hot climate regions. In this paper, the viability of jet impingement cooling for PV panels subjected to the climatic conditions of Dhahran region of Middle East has been discussed. Experimental and numerical performance evaluation was carried out for two configurations, an uncooled PV system and an impingement cooled PV system. The complete cooling model incorporates optical, radiation, thermal, geometric and electrical model for the overall performance analysis of the PV system. Jet impingement geometric model for PV was developed to carry out the heat transfer analysis for single nozzle for analyzing the cell temperature, power output and conversion efficiency of PV strings. For precise prediction of the performance of the PV cell, seven parameters electrical model is employed, whereas for absorbed radiation calculation isotropic sky model is adopted. Temperature measurements revealed cell temperature as high as 69.7 °C and 47.6 °C for an uncooled system for June and December, respectively. By applying jet cooling, average cell temperature was reduced to 36.6 °C for June and 31.1 °C for December. Power output and conversion efficiency was enhanced by 51.6% and 66.6% by employing jet cooling for June, respectively. Similarly, December results revealed performance improvement by 49.6% in power output and 82.6% in conversion efficiency. Simulation carried out reveals that jet cooling with optimum parameters has not only significantly increased the electrical power output and cell efficiency but has reduced the cell temperature while keeping it uniform for each cell.

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1. Introduction

Thermal management of photovoltaic (PV) systems plays a key role for low and high concentration systems. The overall performance of solar panels is strongly dependant by the affect of PV cells' operating temperature. Commercially existing PV modules generally converts 13–20% of incident solar radiation to electricity (Usama Siddiqui et al., 2012). The remaining component of radiation absorbed in the cell transforms into thermal energy which causes the temperature of the cell to be increased. This further causes the panel efficiency to decrease as the open circuit voltage and fill factor is reduced with rising temperature (Amin et al., 2009; Bahaidarah et al., 2013; Du et al., 2013; Erdil et al., 2008; Kumar and Rosen, 2011; Ling et al., 2014; Micheli et al., 2013; Radziemska and Klugmann, 2002; Radziemska, 2003; Teo et al., 2012; Yadav et al., 2013). With an increase in the PV cell temperature, the open circuit voltage and fill factor is reduced, which decreases the conversion efficiency of the PV system (Royne et al., 2005). Hence, the cooling technique is desirable to maintain the cell at the operating temperature and should be such that it keeps the average cell temperature at its minimum with a uniform distribution (Bahaidarah et al., 2014; Chenlo and Cid, 1987; Du et al., 2013; Royne et al., 2005).The temperature distribution in PV panels generally depends on the PV cell material, PV cell type, the panel configuration, the electrical load attached to the PV system, the prevailing environmental conditions and in case of cooling, the characteristics of the heat dissipation technique (Baig et al., 2012). The main reasons for non-uniform flux distribution this issue are the PV cell material, geometry of the concentrator, optical properties of concentrators, design, and manufacturing method (Baig et al., 2012).

In this paper a cooling technology is presented to investigate the optimum geometry which would result in the best global performance with uniform temperature profile and low cell temperature so that the entire string could be operated under similar temperature values to avoid current mismatching and hotspot problem. In a non-uniform temperature distribution, PV cell in a string operates at different power points and results in a faulty mode thereby reducing the performance of PV with reduced efficiency (Gokmen et al., 2013) and increased thermal stresses

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Nomenclature

a	modified diode ideality factor (V)
A_c	area of PV string (m^2)
C_d	discharge coefficient of nozzle
CPV	concentrated photovoltaic
d	diameter of nozzle
e	elementary charge, 1.60218×10^{-19} (C)
F	Reynolds number parameter in jet model
G	total horizontal radiation (W/m^2)
G_b	horizontal beam radiation (W/m^2)
G_d	horizontal diffuse radiation (W/m^2)
G_{ref}	reference condition (STC) incident radiation (W/m^2)
h	heat transfer coefficient (W/m^2 K)
h_w	wind heat transfer coefficient (W/m^2 K)
G_T	total tilted incident radiation (W/m^2)
I	PV output current (A)
I_L	light current (A)
I_o	diode reverse saturation current (A)
k	thermal conductivity, (W/m K)
K	extinction coefficient
$K_{\tau\alpha,b}$	incidence angle modifier for beam radiation
L	thickness of front glass layer (m)
m	irradiance dependence parameter for I_L
M	air mass modifier
N	number of nozzles
n	temperature dependence parameter for ideality factor
Nu	Nusselt number
P	electrical power (W)
P_{net}	net power (W)
P_{pump}	pumping power (W)
ΔP	pressure drop (Pa)
q	heat transfer (W)
Q	flow rate (m^3/s)
r	equivalent radius of cooled plate
R_{beam}	geometric factor for beam radiation
R_s	series resistance (Ω)
R_{sh}	shunt resistance (Ω)
$RMS\%$	root mean square percent deviation (%)
R_{top}	top loss thermal resistance (K/W)
R_{bottom}	bottom loss thermal resistance (K/W)
Re	Reynolds number
S_{pv}	absorbed solar radiation at operating conditions (W/m^2)
S_{ref}	absorbed solar radiation at STC (W/m^2)
T	temperature (°C)
T_{amb}	ambient temperature (°C)
S	plane-of-array absorbed solar radiation at operating conditions (W/m^2)
$T_{f,in}$	inlet water temperature (°C)
T_{cell}	cell temperature (°C)
V	voltage (V)
V_w	wind speed (m/s)
z	nozzle to plate spacing

Greek symbols

β	slope of PV string
δ_w	uncertainty of variable "W"
η_{mp}	electrical efficiency of PV at maximum power point
ρ	reflectivity of ground
θ	incidence angle of solar radiation
θ_r	refracted angle of solar radiation
σ	Stefen–Boltzmann's constant
$(\tau\alpha)$	Transmitivity–Absorptivity product

Subscripts

0	reference condition
al	aluminum
amb	ambient
avg	average
b	beam radiation
bs	backsheet
c, cell	PV cell
conv	convection losses
diff	diffuse radiation
exp	experimental
mod	modeled
martin	martin model
mp	maximum power point
n	normal
oc	open circuit point
opt	optimum condition
ref	reference cell condition
rad	radiation losses
sc	short circuit point
string	parameter for PV string
tot	total
uniform	uniform temperature distribution

(Royne et al., 2005). Moreover this effect further aggravates the performance of PV systems in the case of concentrated systems where the effect of non-uniformity is much pronounced. Non-uniform temperature across the PV string has been found to affect the cell efficiency and overall system performance in a negative way (Baig et al., 2012; Chenlo and Cid, 1987; Luque et al., 1998; Mathur et al., 1984; Royne et al., 2005; Sanderson et al., 1980). Non-uniformity in temperature distribution affect the PV system performance in two ways: (1) Cells experience efficiency loss due to loss in power output; (2) Temperature variation induces thermal fatigue because of large amount of thermal cycles and stresses. This further cause the irreversible damage to solar cell due to excess localized heating across the region and reduces the reliability of the system (Royne et al., 2005). Due to the non-uniform temperature distribution, hot spots occurs on the cell resulting in reduction of life and safety of PV systems (Royne et al., 2005).

Series connected PV cells faces greater damage as current directly varies with light, so the current in an array of series of identical PV cells will be restricted by the solar cell with the maximum temperature. This failure is called as the 'current matching problem'. This issue can be solved by using bypass diodes or by keeping a uniform temperature across each series connection. Moreover conventional cooling mechanism like rectangular channel cooling in simple PVT systems result in temperature variation along the length of PV string because of the effect of constant heat transfer increase at the wall. Therefore, the cooling method should be such that it keeps the average cell temperature at its minimum with a uniform distribution (Baig et al., 2012; Royne et al., 2005).

Therefore the viability of jet impingement cooling, with its high heat transfer capability and uniform cooling characteristics, has been considered in this study to avoid the failure modes of current mismatching and structural damage due to thermal stresses. The proposed study is full scale real time PV setup which has been tested under climate conditions of Dhahran and is an extension from the laboratory scale setup developed by Royne and Dey (2007). Moreover, the PV system model developed by coupling five sub-models shows a good agreement between experimental and simulation results. Electrical model employed is seven parameters model which estimate precise module parameters even for low irradiance and high temperature (Siddiqui and Abido, 2013).

Cooling systems for performance improvement of PV may be classified by the cooling fluid used, concentration of systems (high or low), heat transfer capabilities and temperature characteristics (Bahaidarah et al., 2016). Development of high heat transfer cooling technologies, such as microchannels, spray cooling and jet impingement, have resulted in higher power percentage improvement than corresponding temperature difference based on power temperature coefficients. Abdolzadeh and Ameri (2009) investigated spray cooling and found 48% improvement in power when temperature was reduced from 58 °C to 37 °C. Heat transfer enhancement using nanofluids have shown power improvement by 37.6% where the temperature change was only 20 °C (Karami and Rahimi, 2014). Valeh-e-sheyda et al. (2013) showed 30.75% increase in maximum power of cell at a temperature of 52 °C by applying two phase cooling. Table 1 shows the comparative analysis between cooling techniques for high and low concentration systems with their temperature characteristics.

In the present work, numerical and experimental investigation of jet impingement cooling for PV system was performed. The advantage of using impingement cooling can result in low average cell temperature for PV cell with uniform temperature distribution across series connection. For the comparison, an uncooled PV system was modeled and fabricated as a base case and the experimental test facility was developed to examine the effect of the temperature on the performance of the PV systems. Experimental measurements and analysis for two days of June and December are presented in this paper. Finally the cell temperature, power, and efficiency are compared with an uncooled PV string.

2. Modeling and simulation

Modeling and simulation for the PV string with an uncooled system and an impingement cooling system for PV strings subjected to the climate of Dhahran was performed using Engineering Equation Solver (EES) (Klein and Alvarado, 2015). EES is an algebraic and differential equation solver developed by F-Chart at University of Wisconsin. The main advantage of using EES over other solvers is its built-in mathematical and thermopyhysical property functions, that can be directly employed in the model. Moreover, EES has the capability to do parametric analysis by choosing parameters in a spreadsheet-like table making it user friendly for any engineering related problems.

Fig. 1 shows the schematic for PV system and its sub-models employed for modeling thermal and electrical characteristics. Main input parameters of climate conditions are used an input to optical and radiation model to calculate the absorbed PV radiation. This absorbed PV radiation, wind speed and ambient temperature is then provided to Thermal model coupled with Jet Nozzle parameters to calculate the temperature of the cell. Electrical model uses the module data and cell temperature to estimate the power output and efficiency of the devices. The complete PV performance model for the cooling system incorporates the following sub models:

- Radiation model;
- Optical model;
- Thermal model;
- Geometric model; and
- Electrical model.

Firstly, impingement cooling model was developed and parametric study was carried out to find the optimum heat transfer coefficient for cooling based on minimum cell temperature and maximum power output by the PV. Secondly, based on the optimum heat transfer coefficient, performance assessment of the PV systems using detailed system model was carried out.

The details of the modeling and analysis are discussed in the following subsections. For this study, radiation, optical, geometrical, thermal and electrical modeling were carried out using EES software.

Table 1
Summary of cooling techniques reviewed.

Researcher	Type of cooling system	Nature of work	Type	Coolant type	Cell temperature characteristics
Baloch et al. (2015)	Improved Heat Exchanger as converging channel	Numerical Experimental	Non-CPV	Water	Temperature as high as 71.2 °C for uncooled PV. Reduced to 45.1 °C by cooling Uniform cooling achieved with 0.91 °C standard deviation on the surface of PV
Akbarzadeh et al. (1996)	Heat Pipe cooling	Experimental	CPV 20 X	–	46 °C temperature after cooling
Anderson et al. (2008)	Heat Pipe cooling with fins	Numerical Experimental	CPV	–	40 °C temperature higher than ambient temperature. Temperature profile uniform
Qifen et al. (2013)	Heat Pipe cooling with fins	Numerical	CPV	–	Temperature reduced to 32 °C. Maximum temperature non-uniformity across cell was 3 °C
Liu et al. (2011)	Immersion Cooling	Experimental	CPV	Dielectric Liquid	Cell temperature 30 °C. Uniform Temperature variation of 3 °C
Zhu et al. (2011)	Immersion Cooling	Experimental	CPV 250 X	Dielectric Liquid	Temperature reduced to 45 °C. Maximum temperature non-uniformity across cell was about 5 °C
Han et al. (2011)	Immersion Cooling	Numerical	CPV	Dielectric Liquid	Solar cell temperature reduced to 29 °C using DI liquid
Xiang et al. (2012)	Immersion Cooling with fins	Numerical	CPV 250 X	Dielectric Liquid	For optimum design, maximum cell temperature was 40 °C. Temperature non-uniformity reduced from 15 °C to 10 °C
Min et al. (2009)	Heat Sink	Experimental	CPV 400 X	–	Average cell temperature of 37 °C measured
Huang et al. (2004)	Phase Change Material cooling	Analytical	Non-CPV	–	Temperature maintained in the range of 33–36.4 °C Uniform temperature distribution
Hasan et al. (2010)	Phase Change Material cooling	Experimental	Non-CPV	–	18 °C temperature reduction on front surface
Maiti et al. (2011)	Phase Change Material cooling	Experimental	Non-CPV	–	Panels temperature reduced by 20% For outdoor condition, 62 °C whereas 65 °C for indoor was achieved
Royne and Dey (2007)	Hydraulic Impingement Cooling	Analytical	CPV 200 X and 500 X	Water	For 200 X, cell temperature decreased from 60 to 30 °C. For 500 X, cell temperature decreased from 110 to 40 °C
Barrau et al. (2014)	Hybrid Microchannels–Impingement	Experimental	CPV 1905 X	–	Temperature variation of 2.01 °C was obtained for micro-channel heat sink For hybrid cooling standard deviation of 0.46 °C was obtained

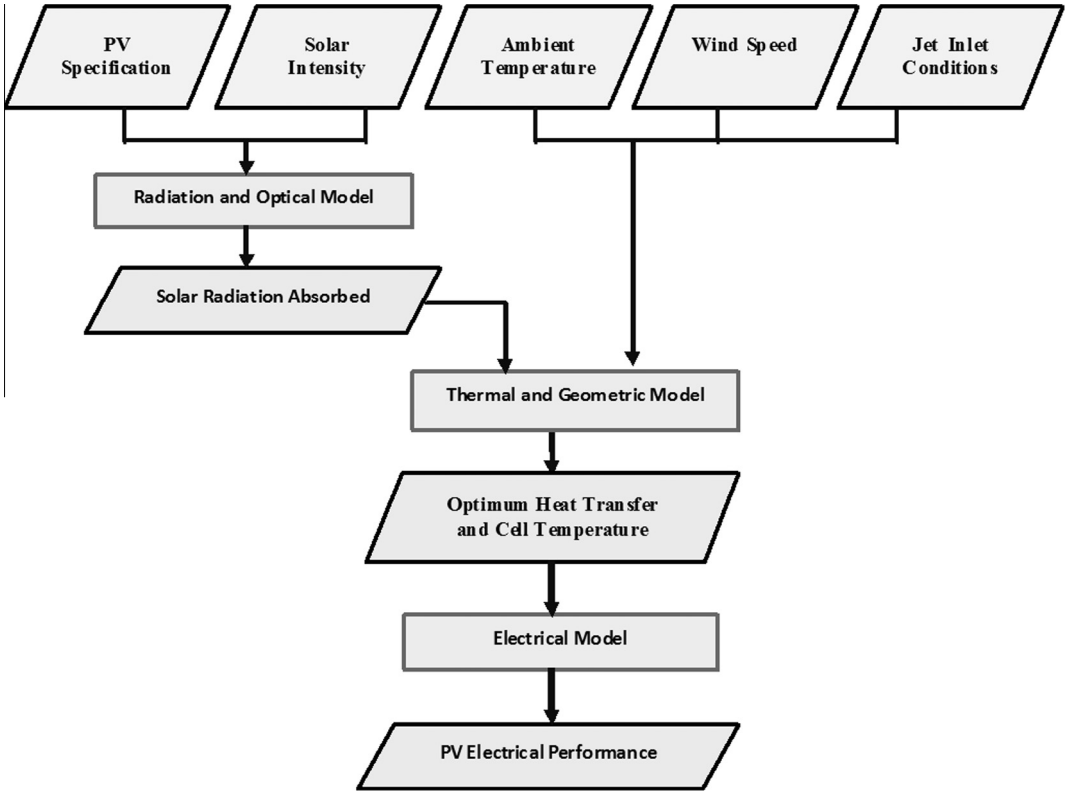


Fig. 1. Flowchart of relations between sub-models of PV systems.

2.1. Optical and radiation model

Diffuse radiation model employed for the analysis was the isotropic sky model (Liu and Jordan, 1961) which considers all components of radiation for calculating the incident radiation on the tilted plane whereas optical model was used for calculating the solar radiation absorbed in the PV cell. The absorbed radiation, denoted by S_{PV}, contains direct, diffuse and reflected irradiance and is calculated from the radiation and optical model which is then coupled with the electrical model and the thermal model to determine the electrical power output and cell temperature. The product of transmittance absorbance product of solar cell was calculated separately for beam, diffuse, and ground reflected radiation. Subscript "b" denotes beam and "d" denotes diffuse radiation whereas subscript "g" denotes ground reflected.

Absorbed radiation, S_{PV}, based on isotropic model for PV cell is given by (1) (Liu and Jordan, 1961)

$$S_{PV} = S_{ref}\left[M\frac{G_b}{G_{ref}}R_b K_{\tau\alpha,b} + M\frac{G_d}{G_{ref}}K_{\tau\alpha,d}\frac{1+\cos\beta}{2} + M\frac{G}{G_{ref}}\rho K_{\tau\alpha,g}\frac{1-\cos\beta}{2}\right] \quad (1)$$

$$S_{ref} = (\tau\alpha)_n G_{ref} \quad (2)$$

where R_b is the geometric factor and subscript *ref* denotes reference conditions. S_{ref} is calculated by using Eq. (2) where G_{ref} denotes the solar radiation at a reference condition of 1000 W/m^2 (Duffie and Beckman, 2013). While determining the $(\tau\alpha)$ product, internal reflection losses are also incorporated which occur in the different layers of the solar cell. Air mass modifier, M, is calculated using (3) for monocrystalline solar cells described by (King et al., 2004).

$$M = \sum_0^4 a_i(\mathrm{AM})^i \quad (3)$$

where AM is the air mass and the a_i are constants for PV cells. For monocrystalline PV, the values of a_0, a_1, a_2, a_3 and a_4 are 0.935823, 0.054289, −0.008677, 0.000527, and −0.00001.

2.2. Geometric model

The aim of geometric modeling was to find the optimum jet parameters to ensure low temperature of solar cell. This will result in the maximum possible net power output by the jet cooled PV panels. The fluid dynamics and heat transfer characteristics for jet impingent are dependent on the following parameters: dimensionless nozzle to plate spacing z/d, nozzle shape, prandtl number, nozzle diameter and number of nozzles. Fig. 2 shows the key

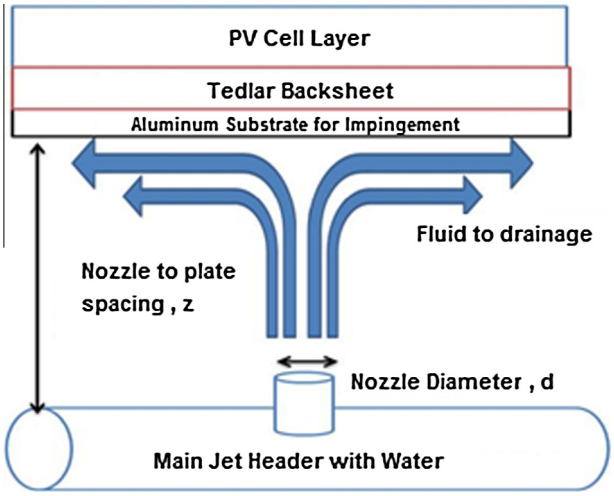


Fig. 2. Schematic for geometrical parameters of jet cooling model for PV.

parameters involved in jet impingement cooling with the main parameters highlighted in Table 2.

Tie et al. (2011) found an optimum of value of dimensionless nozzle to plate spacing $z/d = 5$ by incorporating the correlation into their model. Martin (1977) in optimizing the work by Krotzsch (1968), showed an optimum z/d of 5.43. Long pipe nozzles were used to ensure fully developed flow at the nozzle exit. The length of the potential core (in units of nozzle diameter) was shown to increase with increasing nozzle diameter.

For single nozzle system of diameter (d) and fluid's Prandtl number (Pr), model employed by Royne and Dey (2007) was used by employing Eqs. (4) and (5). F (dimensionless parameter) relates Reynolds number (Re) to Average Nusselt Number (Nu_{avg}) by Eq. (5). This correlation used various nozzle configurations which make it suitable for any nozzle type. The experimental limits for this correlation is from, $Re = 2000–400{,}000$, z/d: 2–12, r/d: 2.5–7.5. (z/d) opt = 5.3 was found to be an optimum nozzle to plate spacing.

$$F = 2Re^{0.5}\left(1 + \left(\frac{Re^{0.55}}{200}\right)\right)^{0.5} \tag{4}$$

$$Nu_{avg} = \left(\frac{(1 - 1.1 \cdot (d/r))}{1 + 0.1 \cdot ((z/d) - 6) \cdot (d/r)}\right) F(d/r) Pr^{0.42} \tag{5}$$

2.3. Thermal modeling

Temperature characteristics on the PV surface are affected by cooling technique, solar intensity, radiation absorbed, convection losses by wind, radiation losses, wind speed, type of material and length of the layer of material. The absorbed radiation (S_{pv}) calculated along with cell parameters and PV site information is thus given as an input to the thermal model to calculate the cell temperature. Thermal modeling for impingement cooling is developed using an energy balance approach (Royne and Dey, 2007; Sarhaddi et al., 2011) and by applying thermal resistances at the nodes of different layers of PV string and cooling setup. For the case of an uncooled PV panel, modeling is carried out using the thermal network diagram (Sarhaddi et al., 2011). Water is used as a cooling fluid in the channel. Table 3 shows the properties of the material used in the modeling and the simulation of the cooled and uncooled system. Fig. 3 shows the schematic of different layers of PV systems with impingement cooling and without cooling with respective thermal networks. By applying energy balance approach on the PV system and solving equation at the nodes for different layers, temperature characteristics of the PV string was obtained.

Fig. 3 shows the thermal network diagram of the uncooled PV string comprising on the layers of the solar cell and tedlar back sheet. Thermal resistance network diagram was developed and simultaneous equations were solved using EES.

Based on above assumptions and energy balance approach, a thermal model is developed to predict the heating and electrical characteristics of the PV panel. Absorbed radiation estimated is employed to evaluate the thermal model for the temperature distribution within the layers of the impingement cooled PV system. For jet cooling system, optimization has been carried out on the basis of pumping power so that the optimum parameters could be used for PV panel modeling.

Energy balance equations for the case of jet impingement PV system are written as follows

$$q_{loss} = S_{PV} - P_{mp} \tag{6}$$

$$q_{loss} = q_{bottom} + q_{top} \tag{7}$$

$$q_{top} = \frac{T_{cell} - T_{amb}}{R_{top}} \tag{8}$$

$$q_{bottom} = \frac{T_{cell} - T_{amb}}{R_{bottom}} \tag{9}$$

When designing a jet impingement device, it is not only the flow rate which is important, but also the pressure drop through the device. The preferred cooling system will in many cases be the one that delivers the highest rate of cooling at a given pumping power. The total pumping power is proportional to the product of flow rate and pressure drop.

Pressure drop across jet nozzles can be found using relation (Royne and Dey, 2007) and then calculating net power developed by the system utilizing pump power (P_{pump}).

$$\Delta P = \frac{8 \cdot \rho \cdot Q^2}{(3.142 \cdot C_d \cdot d^2)^2} \tag{10}$$

$$P_{pump} = \Delta P \cdot Q \tag{11}$$

Utilizing the energy balance approach on the layers shown in Fig. 3 of the PV system and analyzing the thermal network, the following equations are simultaneously solved to obtain the temperature characteristics of the PV string. For the thermal modeling of the uncooled PV system, similar analysis was performed without considering the aluminum substrate for impingement and jet cooling resistance in the thermal network.

2.4. Electrical modeling

The electrical model used in this analysis is a seven parameters PV equivalent model presented by Siddiqui and Abido (2013). This model gives highly accurate and precise result while using data from the module sheet provide by the manufacturer.

The estimation of five parameters for the single diode model of PV requires complex techniques because of the extremely nonlinear character of the current voltage relationship. The seven parameters model is solved using optimization techniques based on the minimization of errors at high temperature and low intensity module data points. The basic five parameters that governs the IV characteristics of the PV panel are light current "I_L", reverse saturation diode current "I_o", ideality factor "a", series resistance "R_s", and shunt resistance "R_{sh}".

$$I = I_L - I_o \cdot \left(\exp\left(\frac{V + I \cdot R_s}{a}\right) - 1\right) - \frac{V + I \cdot R_s}{R_{sh}} \tag{12}$$

Table 2
Overview of jet cooling parameters considered.

Cooling parameter	Details
Nozzle diameter, d	5 mm
Dimensionless Nozzle to plate spacing z/d	5.3
Number of nozzles, N	1
Prandtl number, Pr	5.1
Cooling fluid	Water
Nozzle shape	Pipe

Table 3
Properties of materials used for cooled and uncooled PV string in thermal analysis.

PV system layer	Material	Conductivity (W/m K)	Specific heat (J/kg K)	Density (kg/m^3)
Solar cell	Silicon	148	677	2330
Back sheet	Tedlar	0.033	1250	1200
Impingement Substrate	Aluminum	270	903	2702

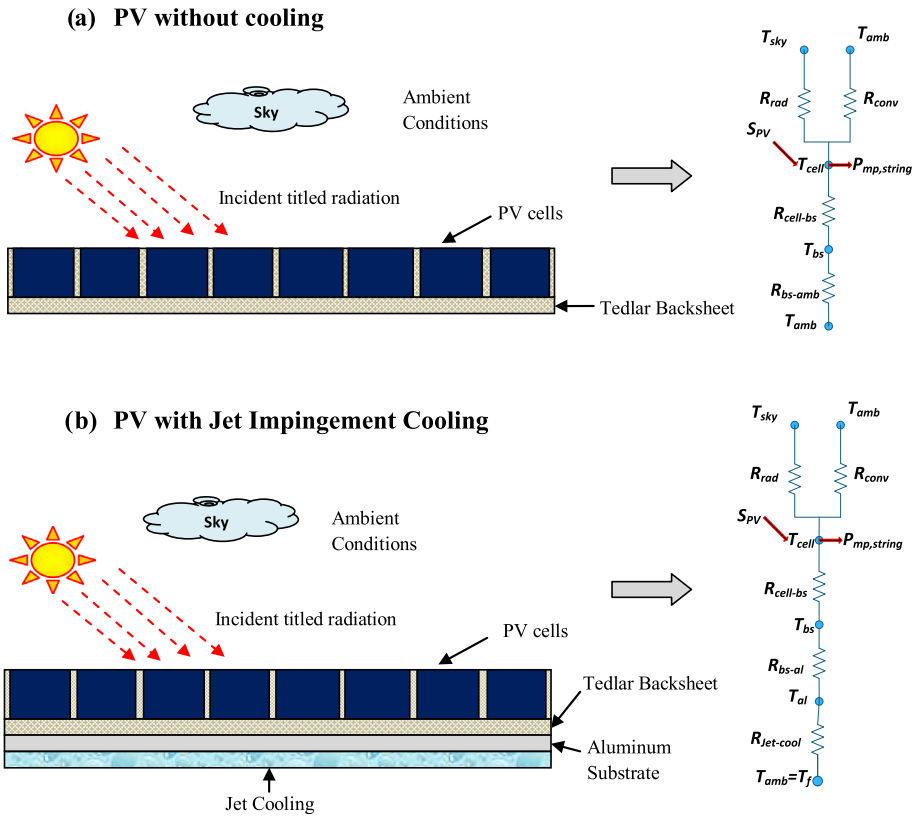


Fig. 3. Cross sectional view and thermal network diagram of a PV string (a) without cooling and (b) with impingement jet cooling.

where I and V are the current and voltage at a given load.

The following conditions are used to estimate the five reference parameters:

At short circuit	$[dI/dV]_{sc} = -1/R_{sh,ref}$
At open circuit voltage	$V = V_{oc,ref}$, $I = 0$
At short circuit current	$V = 0$, $I = I_{sc,ref}$
At the maximum power point	$V = V_{mp,ref}$, $I = I_{mp,ref}$
At the maximum power point	$[dI/dV]_{mp} = 0$

The two new introduced variables in the seven parameters model are modified translating parameters "m"&"n" to convert reference conditions to operating conditions as shown by Eqs. (13) and (14). These parameters increase the precision of the PV electrical parameters for high cell temperature and low irradiance.

After obtaining the five parameters, the remaining two variables are found by,

$$a = a_{ref} \cdot (T_{cell}/T_{ref})^n \tag{13}$$

$$I_L = (S_{PV}/S_{ref})^m \cdot (I_{L,ref} + \mu_{isc} \cdot (T_{cell} - T_{ref})) \tag{14}$$

$$error = \sqrt{(P_{\mathrm{mod}} - P_{\exp})^2}|_{Low\ Irradiance} + \sqrt{(P_{\mathrm{mod}} - P_{\exp})^2}|_{High\ Temperature} \tag{15}$$

The objective function shown by Eq. (15) is used for error minimization and was calculated using power at low irradiance and high temperature using MATLAB software. In this present study EES is used to evaluate these five parameters at reference conditions $T_{cell,ref} = 25\ ^\circ C$, $I_{ref} = 1000\ W/m^2$.

The maximum power output, efficiency and net power can be obtained as.

$$P_{mp} = I_{mp} V_{mp} \tag{16}$$

$$\eta_{mp} = \frac{P_{mp}}{A_c \cdot G_T} \tag{17}$$

$$P_{net} = P_{mp} - P_{pump} \tag{18}$$

3. Experimental setup and uncertainty

3.1. Experimental setup

Experiments were conducted for several hot weather days during June and cold weather days during December between 9:00 AM and 3:00 PM. Data collection from the experimental setup was carried out to investigate the performance of electrical efficiency and power output from the PV strings during operation under normal concentration. The PV strings employed were manufactured by connecting eight solar cells in series with bottom contacts. Maximum Power Point Tracker (MPPT) device was used during operation to extract maximum current and maximum voltage. The electrical energy generated by the module was stored in 12 V batteries. In order to measure the temperature on the PV surface, eight Type K calibrated thermocouples were used and the temperature profiles were recorded using data acquisition system.

Calibrated thermocouples of type K were used to measure average cell temperature and temperature distribution on the surface of the PV strings which was directly monitored on Data Acquisition (DAQ) system. The DAQ system installed for this research was capable of monitoring and storing calculations of the voltage, current, wind speed, ambient temperature, light intensity, temperature, and maximum power. All the panels were aligned along east west direction and facing south. Inclinations of the panels were adjustable for optimization of the collectible solar radiation in different

seasons. Solar intensity was measured using pyranometer on the same plane as the PV string whereas anemometer was used for measuring the wind data. Optimized flow rate of 1.5 L/min for water was used for fluid circulation in the jets after preliminary experimental investigations to ensure the characteristics of jet. The measurements for the performance parameters were recorded for several days and for the purpose of briefness two days data, one for summer and one for winter, has been analyzed in the present study.

The aim of the jet cooling was to design and build a completely self-contained impingement test rig for the cooling of PV string. This was achieved by single jet exposed to each cell such that the average temperature of each cell remains constant and adjustable stand. The main design consideration was the nozzle diameter, nozzle shape, flow rate, nozzle to plate spacing and the tilt angle of the jet to obtain approximately uniform temperature profile through the cell.

Fig. 4 shows the shows the layout of jet cooling system for PV string and the location of. The main design issue was the main frame. This was manufactured using plexiglass so that the physics of jet impingement can be monitored and nozzle locations can be modified accordingly. Strainer was used before main water inlet to ensure that no clogging takes place. A pump and bypass system was used in order to maintain the pressure of water supplied to the plenum. Based on the literature, modeling simulations, and impingement jet test rig optimum parameters were chosen for the manufacturing of jet impingement design for the experimental purpose. Entire design was performed with leak proofing in mind.

Tables 2 and 3 summarizes the important parameters considered in this study. Fig. 4 shows the basic manufacturing cycle of impingement cooling setup for PV and the final setup installation on site at the test facility.

3.2. Uncertainty

An uncertainty analysis for all the test instrumentations involved in the measurements of the data was performed. Accuracies and ranges of the instrumentations were utilized to investigate the propagation of uncertainty on variable to another. The experiments were performed outside in the actual environment and bias errors were taken into account when formulating the analysis for uncertainties.

Table 4
Uncertainty values of an equipment.

Equipment	Uncertainty value (δ_W)
Thermocouple	0.5 °C
Solar intensity	10 W/m^2
Wind speed	0.28 m/s
Ambient temperature	0.1 °C
Current	0.059 A
Voltage	0.00086 V
Power	0.27 W

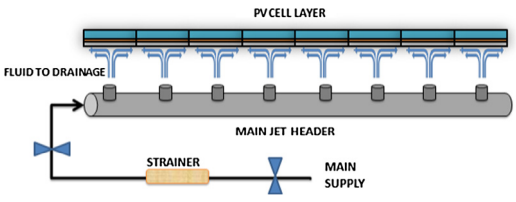
Kline and McClintock (1953) developed a method to calculate the bias error of calculated variables. For a calculated variable W, which is a function of n variables v_i, the uncertainty of W, δ_W can be found by

$$\delta_W = \left[\left(\frac{\partial W}{\partial v_1}\delta v_1\right)^2 + \left(\frac{\partial W}{\partial v_2}\delta v_2\right)^2 + \ldots + \left(\frac{\partial W}{\partial v_n}\delta v_n\right)^2\right]^{\frac{1}{2}} \quad (19)$$

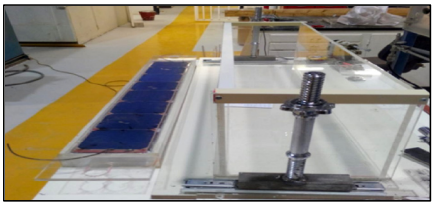
Using the above formulation and the ranges provided by the instrument manufacturers, uncertainty values for all the instrumentations are listed in Table 4.

4. Results and discussion

In this section the experimental and numerical results are discussed for various parameters which determine the overall performance of the uncooled and jet impingement cooled photovoltaic systems. The main input parameters for the models are the meteorological data in the form of ambient temperature, wind speed, incident radiation, flow rate of coolant, inlet temperature of the fluid and solar cell parameters. From these data the system model is capable of evaluating the performance characteristic of the PV panel. The results from thermal model were given input to the electrical model to calculate the conversion efficiency and power output. All these input variables have been collected at Dhahran, latitude 26.5°N, for the month of June and December 2014. Jet impingement cooling parameters are evaluated first using standard testing conditions for Dhahran and optimum conditions are found. After the selection of jet parameters from modeling and lit-



Schematic of Jet Impingement Cooling Setup



Fabrication of Plexiglass frame and PV String Installation.



Jet Impingement Test Rig Installed on Site



Adjustable stand for PV string

Fig. 4. Test apparatus of jet impingement test rig for PV and manufacturing cycle.

erature, thermal and electrical characteristics are examined in the subsequent sections.

Experimental measurements and analysis for two days of June and December are presented in this paper. For the purpose of validation of the results from modeling and experiments, a root mean square percentage deviation (RMS %) has been evaluated using the following equation (Bahaidarah et al., 2013; Joshi et al., 2009):

$$\text{RMS}\ \% = \sqrt{\frac{\sum [100 \times (X_{sim,i} - X_{\exp,i})/X_{sim,i}]^2}{n}} \quad (20)$$

where n is the number of experiments.

Table 5 shows the variation of the ambient temperature, wind speed and solar intensity collected from the site for analysis. The ambient temperature for the month of June ranges from 33.33 °C to 48.83 °C showing hot conditions whereas ambient temperature drops to the range of 23.6–26.07 °C showing cold conditions for December. Incident Radiation as shown in Table 5 is also higher in the month of June with clear skies and sunny days whereas for December the light intensity was as low as 407 W/m^2 because of winter conditions.

4.1. Jet cooling parameters modeling

Jet modeling simulations are first performed for the parameters to be selected for impingement cooling of PV. Fig. 5 shows the effect of cooling on the performance of cell temperature and power produced by PV string. It is evident that there exists a "knee point" in these curves, after which there would be insignificant improvement in the jet cooling model. Parametric study was carried out using conditions mentioned in Fig. 5 based on the minimization of cell temperature and maximization of power output by PV string. The study showed that after heat transfer coefficient of 1000 W/m^2 K, the difference between consecutive measurements is very small i.e. 0.01 W for the case of power output for the current set of conditions. Based on this result, average heat transfer coefficient of 1000 W/m^2 K is used in the modeling of this study for jet. At this average heat transfer coefficient, selection of jet nozzle diameter is also carried out.

Based on literature and the fabrication issues, dimensionless nozzle to plate spacing (z/d) was selected to be 5.3. For the present study, long pipe nozzle diameter of 5 mm was selected based on the net power analysis, maximum coverage for the cell, jet characteristics, and feasibility of fabrication. It can be seen that there is a significant effect of changing nozzle diameter due to the decrease in pressure drop and frictional losses for small diameter nozzles. It should be noted, however, that pumping power in this study only takes into consideration pressure drop across nozzle.

Table 5
Climatic conditions for test days.

Time	Incident radiation (W/m^2)	Ambient temperature (°C)	Wind speed (m/s)
23rd June climate data			
10:00	864.6	48.83	0.25
11:00	969.2	47.35	2.71
12:00	958.0	42.78	2.83
13:00	889.5	46.43	3.53
14:00	740.5	33.33	4.28
16th December climate data			
10:00	732.6	23.63	1.23
11:00	848.8	24.87	3.40
12:00	674.4	26.07	2.43
13:00	476.7	24.33	1.83
14:00	407.1	25.11	1.69

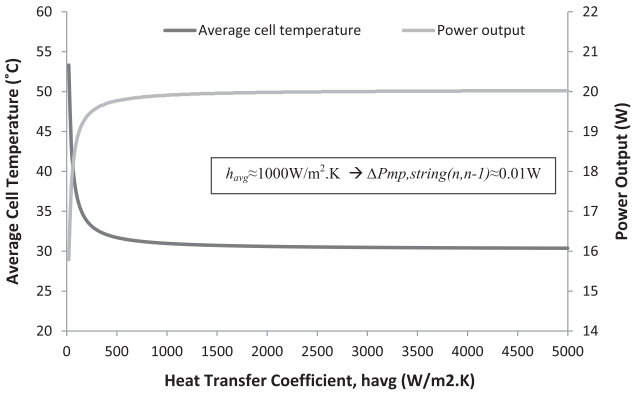


Fig. 5. Effect of heat transfer coefficient of cooling technique on average cell temperature and power output. (G_T = 1000 W/m^2, V_w = 2 m/s, T_{amb} = 37 °C, $T_{f,in}$ = 30 °C). $h_{avg} \approx$ 1000 W/m^2 K is used for jet impingement cooling.

4.2. Optical and radiation modeling

Fig. 6 shows the absorbed radiation by the PV string calculated for the two days considered. As the day progresses the light intensity on the PV increases with an increase in the zenith angle and position of sun. This behavior results in an increase of the absorbed radiation by the PV string and resulting to a maximum value of 795 W/m^2 for June and 708 W/m^2 for December. It is clear that as the day progresses, the solar energy absorbed by the PV string increases reaching to a maximum then declining depending upon the season. The absorbed radiation calculated is used as an input to the thermal and electrical model.

4.3. Thermal modeling and experiments

Experimental measurement and modeling for temperature was performed in order to analyze the performance of the uncooled and jet cooled PV system. Fig. 7 shows the effect of jet cooling on the thermal characteristics of single cell obtained by experiments. Five thermocouples were installed on the surface of PV cell and jet cooling was applied for approximately a period of 4000 s. Temperature non-uniformity is shown on the right axis of the Fig. 7 where it can be seen that after attaining steady state it lies around 1 °C. Also the PV surface temperature profile shows that PV by applying jet cooling, temperature has dropped from 46 °C to 26 °C within 500 s i.e. transient state. In the steady state, it is noteworthy that the temperature of PV surface for all the thermocouples give uniform temperature profile with little variations from the experimental DAQ systems. The temperature uniformity for the case of impingement

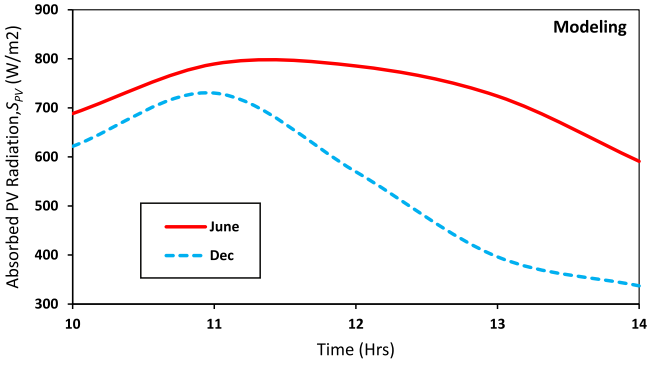


Fig. 6. Absorbed radiation calculated during the test days (simulated data for two days).

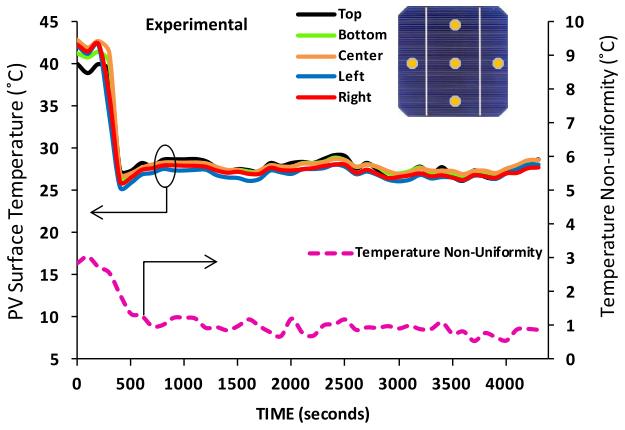


Fig. 7. Single cell surface temperature profile variation with temperature non-uniformity for PV cooling (test conducted on December at 11:00 AM).

cooling results from the incorporation of a single nozzle to cool single cell.

Fig. 8 shows the experimental PV surface temperature along the length of PV string with each thermocouple installed at center of single PV cell. Comparison between PV surface temperature profile for June and December is shown. It is clear the effect of jet cooling in the month of December yielded lower PV surface temperature profile than June. This behavior is due to the lower ambient and water temperature in the winter month. It should be noted that for both seasons, the temperature profile is uniform which shows that jet cooling in extreme climatic conditions can perform better than other cooling mechanisms. However, temperature uniformity is observed to be higher in the month of December than June. For the month of June, PV surface temperature profile showed higher readings varying in the range of 34.5–38.8 °C throughout the day whereas these values were reduced down to the range of 26.8–32.5 °C for the month of December. The variation of temperature readings can also be attributed to the uncertainty of thermocouples and DAQ system.

Fig. 9 shows the variation of average cell temperature during the test days. Comparison between an uncooled PV and jet impingement cooled PV is shown for the two extreme seasonal months selected. It is evident that the application of cooling has shown significant results for the conditions of Dhahran. For an uncooled PV string, maximum temperature of 70.7 °C was reported for June from modeling whereas the maximum modeled cell temperature of 50.1 °C was recorded for December. In comparison to

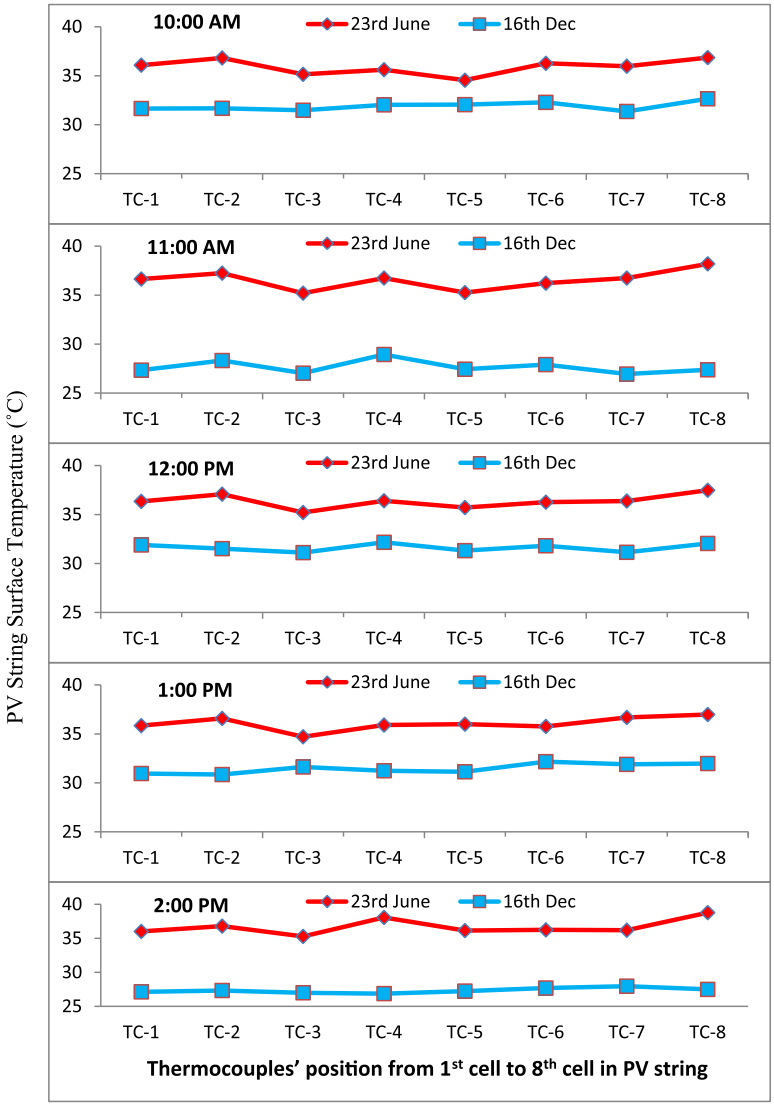


Fig. 8. Experimental PV surface temperature along the length of PV string with thermocouples.

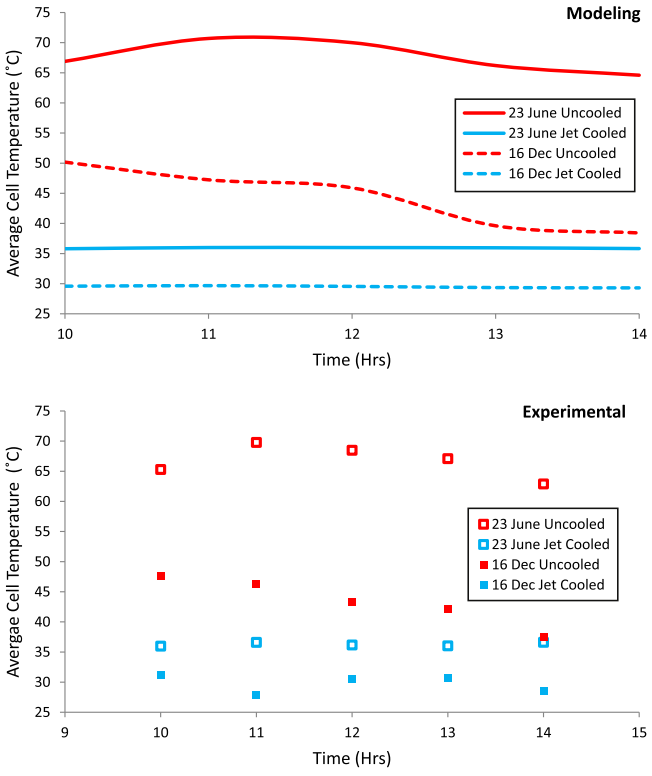


Fig. 9. Variation of average cell temperature for cooled and uncooled PV.

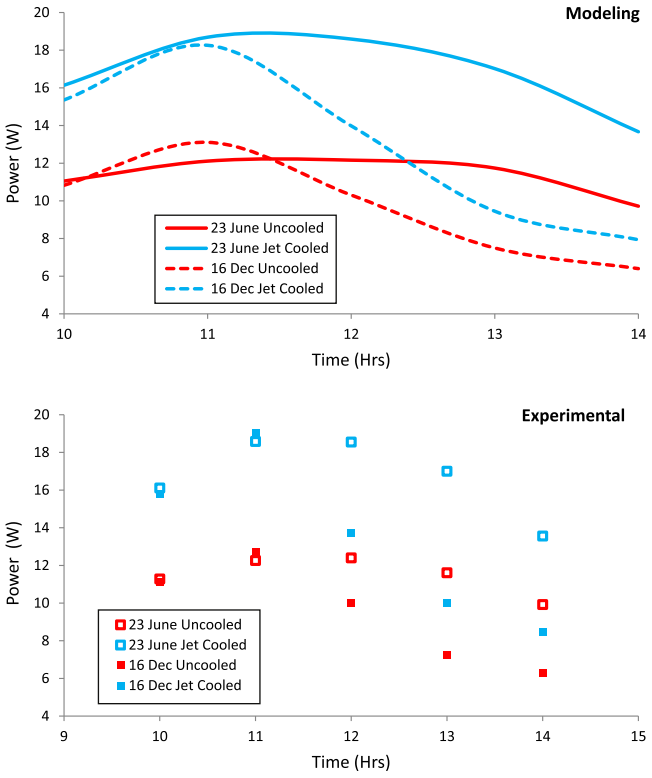


Fig. 10. Variation of power output for cooled and uncooled PV.

this, simulation showed cooling the PV string by resulted in temperature as high as 35.9 °C for June and 29.5 °C for December. Experimentally, maximum temperature of 69.7 °C and 47.6 °C were recorded for uncooled PV for June and December, respectively. By applying cooling in the experimental setup, PV string resulted in a minimum temperature of 36.6 °C for June and 31.1 °C for December. This decrease of average cell temperature is due to the high extraction capability of jet impingement cooling. By using impingement cooling, PV string has been cooled effectively by reducing the cell temperatures while keeping the uniform distribution of the temperature.

In order to compare the performance of cooled PV cell temperature with the base PV system i.e. an uncooled PV, relative percentage reduction was calculated which is defined as:

$$\%T_{cell}reduction = \frac{T_{cell,jet\ \ cooling} - T_{cell,uncooled}}{T_{cell,uncooled}} \times 100 \quad (21)$$

For the month of June, where the effects of high cell temperature is more pronounced, cell temperature experimentally was reduced by 47.5%. In addition, employing cooling in cold weather of December resulted in experimental cell temperature reduction of 34.7%

4.4. Electrical modeling and experimental

The output power generated and the conversion efficiency are the most important parameters for PV system. After using the seven-parameters model described in 2.4, the absorbed radiation and cell temperature was input into the electrical model to analyze the electrical performance of the PV system. The effect of incorporating cooling and comparing it with an uncooled PV system for the months of June and December is shown in Fig. 10. Electrical power output for two test days is shown in Fig. 10 which shows that the use of cooling enhances the performance of the PV for both seasons. Modeling showed that the maximum power output increased from 12.1 to 18.7 W for the case of June while for December, the maximum power output increased from 13.1 to 18.3 W. The experimental results showed similar trend where the maximum power output was increased from 12.4 to 18.8 W whereas for December the increased was from 12.7 to 19.0 W by jet cooling. Comparison of the performance of cooled PV power output with the base PV system i.e. an uncooled PV was performed using:

$$\%P_{mp}increase = \frac{P_{mp,jet\ \ cooling} - P_{mp,uncooled}}{P_{mp,uncooled}} \times 100 \quad (22)$$

For the month of June, PV power output was experimentally enhanced by 51.6% whereas the performance improvement for December was 49.6%. It is important to note here that the PV power output, in addition to cell temperature which is the key parameter, was also affected from incident radiation, solar energy absorbed by PV and wind speed.

Fig. 11 shows the comparison between conversion efficiency of the two systems considered. It is clear from Fig. 11 by employing jet cooling, the performance of the PV has significantly improved. The experimental efficiency increases from 9.3% to 15.5% for the case of June. For December, the experimental efficiency increases from 9.8% to 17.9%.Percentage improvement in the performance was 66.6% and 82.6% for June and December, respectively. Similar trend was followed by the modeling results with an increase in maximum efficiency from 9.2% to 15.5% for June and 10.0% to 17.1% for December, thereby enhancing the overall performance of PV system in both thermal and electrical characteristics. Therefore in order to utilize the maximum benefit from the available PV system, cooling is necessary.

The main factor for different values from experiments and power temperature coefficients is due to the change in climate conditions like operating radiation, wind speed and ambient temperature. The power temperature coefficient tested at lab is reported using Standard testing conditions of 1000 W/m^2

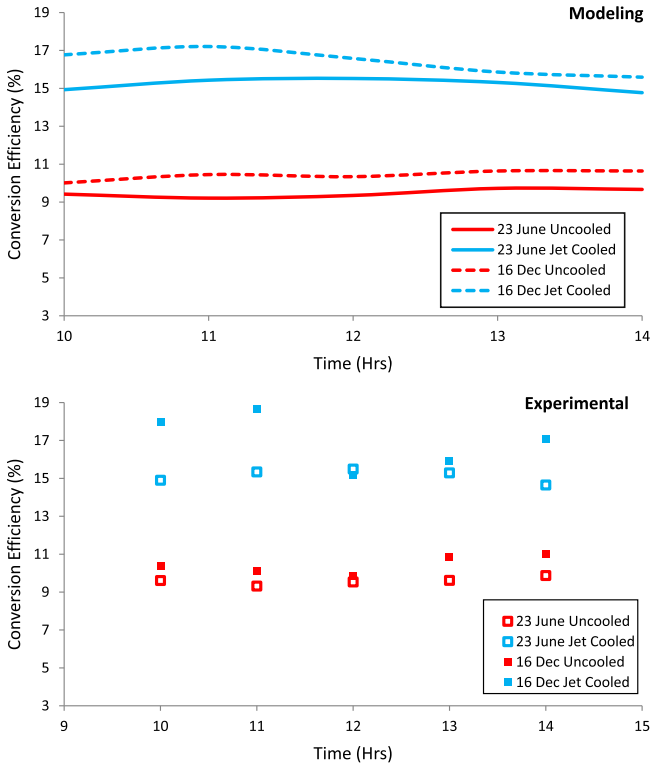


Fig. 11. Variation of conversion efficiency for cooled and uncooled PV.

radiation, 1 m/s wind speed and 25 °C Cell temperature. The highest power, P_{mp}, is the product of two factors, I_{mp} and V_{mp}, both of which differ with temperature and irradiance (King et al., 1997). The discrepancy between reported results and power temperature coefficient is also due to a host of factors, weather variables such as ambient temperature, local wind speed, and solar radiation flux/irradiance, with material and system-dependent properties as parameters, e.g., glazing-cover transmittance (Dubey et al., 2013). Moreover, the compound contributions of the open-circuit voltage and fill factor, changes in the current ratio between subcells with temperature, and variable parasitic series resistance effects in the measurement setup (Kinsey et al., 2008).

4.5. Validation and comparison with other cooling mechanism

In this section the validity of the numerical modeling technique is verified by the experimental results. The numerical results obtained are validated with the experimental data from the tests conducted at Dhahran in the month of June and December, 2014. Figs. 12 and 13 show the comparison of the numerical and experimental results of the average cell temperature and power output for the uncooled and jet cooled PV systems. Fig. 13 demonstrate good agreement; with the root mean square percent deviation 'RMS%' for the experimental and calculated values between 1.7% and 4.8%. Results from the numerical model follow the same trend as measured experimentally. This shows that there is a good agreement between the numerical and experimental results. For the comparison of power output form modeling and experimental, maximum RMS% was found to be between 1.3% and 4.5% showing

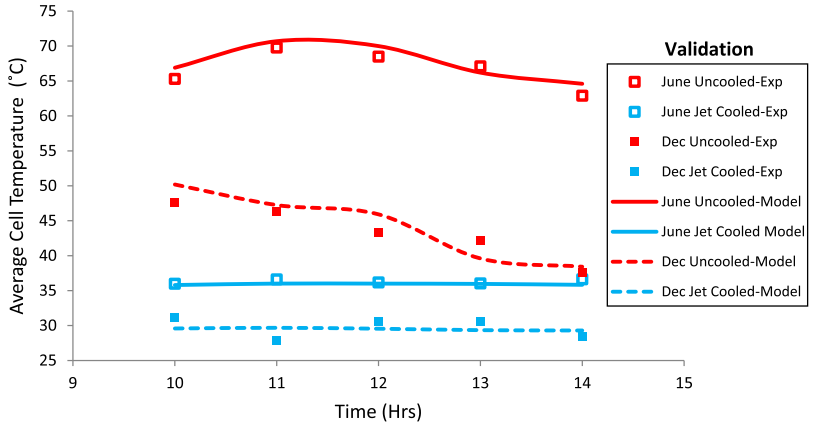


Fig. 12. Validation of average cell temperature from modeling and experiments for cooled and uncooled PV during test days.

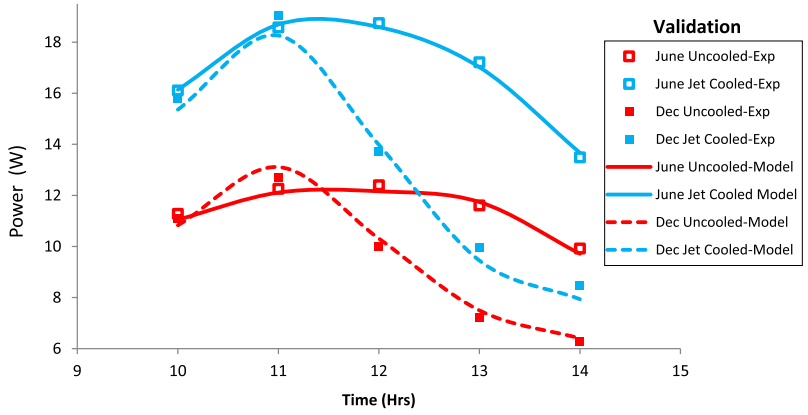


Fig. 13. Validation of power output from modeling and experiments for cooled and uncooled PV during test days.

Table 6
Comparison by simulation of different cooling techniques for PV and an uncooled PV.

16-December	Cell temperature			Power output		
Time	Uncooled	Simple PV/T	Jet	Uncooled	Simple PV/T	Jet
10	50.17	36.65	29.58	10.83	12.55	15.36
11	47.24	38.16	29.672	13.11	15.46	18.26
12	45.91	33.57	29.546	10.31	11.84	13.98
13	39.61	32.64	29.345	7.497	8.21	9.448
14	38.42	31.27	29.298	6.4	7.11	7.933

good agreement. Therefore, the numerical model provided a realistic simulation of the temperature characteristics of cooled and uncooled PV systems and can be a very effective tool for predicting their performance under different operating conditions, enabling us to select the appropriate system for the location considered.

Table 6 shows the highlights the comparison study by simulating different cooling techniques with and uncooled PV. In order to compare temperature and power, two cooling systems were modeled by changing the heat transfer coefficients in the model developed in Section 2. Cooling by Photovoltaic Thermal Collector (PV/T) with simple heat exchanger showed similar trends as jet cooling but with the less order of reduction in temperature. The maximum cell temperature reduction for PV/T was 31.24 °C corresponding to the power output of 7.10 W at 2:00 PM. The high heat transfer coefficient of jet cooling resulted in lower temperature of PVC depending primarily on the inlet water temperature whereas heat exchanger, due to its limited heat transfer capability, resulted in relatively lower performance improvement. It can concluded by Table 6 that jet cooling will perform better than normal cooling. This should be extended to concentration system.

5. Conclusion

The present study deals with the modeling and experimental investigation of jet impingement cooling system for the PV string subjected to the hot climatic conditions of Dhahran. Performance evaluation was carried out for two configurations, an uncooled PV system and an impingement cooled PV system. In the PV string consisting of eight solar cells, each cell was supplied with jet cooling hence resulting in same average temperature, and resolving the issue of current mismatching problem and hotspots. Optical, radiation, thermal, geometric, and electrical models were coupled to assess the overall performance of the PV systems. The results from numerical simulation showed a good agreement with the experimental measurements. The effect of heat transfer coefficient on the performance of PV was studied where the study showed that after heat transfer coefficient of 1000 W/m^2 K, there is insignificant improvement in the performance of PV for the current set of conditions. Based on the analysis, jet cooling parameters were evaluated. Temperature non-uniformity was found to be around 1 °C for single solar cell under jet cooling exposed to hot climate of Dhahran. Temperature measurements revealed cell temperature as high as 69.7 °C and 47.6 °C for an uncooled system for June and December, respectively. By applying cooling, average cell temperature was reduced to 36.6 °C for June and 31.1 °C for December. Temperature profile for jet cooled PV string resulted in a temperature range of 34.5–38.8 °C throughout the day whereas these values were reduced down to the range of 26.8–32.5 °C. Experimentally, maximum temperature was reduced from 69.7 to 36.6 °C and 47.6 to 31.1 °C by employing cooling for June and December, receptively. Using this active cooling technique, cell temperature was reduced by 47.5% for June and 34.7% for December. Power output and conversion efficiency was improved by 51.6% and 66.6% by employing jet cooling for the data of June, respectively. Alternatively, December results revealed performance enhancement by 49.6% in power output and 82.6% in conversion efficiency. Thus, the proposed cooling system could be employed in the extreme climatic conditions for proper operation at low cell temperature with uniform temperature distribution.

Acknowledgment

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